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1.Purpose

This plan defines how Steam Sterilizer S-101 and its control system will be shown, with documented evidence, to be fit for sterilizing wrapped instrument trays (Load configuration LC-01) Using cycle C-121. It sets the approach, responsibilities, deliverables, acceptance criteria, and the reason for the key parameters' values.

2. Scope

In scope: The sterilizer, its control system, cycle C-121, load configuration LC-01, electronic records and signatures, backup and recovery and interfaces to the time server, document management system (DMS), and quality system.

Out of Scope: Facility HVAC and steam generation, packaging design, product-specific sterility assurance studies, and biological indicator studies.

3. System Description

S-101 is pre-vacuum steam sterilizer controlled by PLC with an HMI. Chamber temperature (TT-101) and pressure (PT-101) are recorded for each cycle in database with an audit trail. Details are in part 1.

4. Validation Approach

Validation follows a risk-based lifecycle consistent with ISPE GAMP 5 (Second edition), ISO 17665, and the FDA Quality management System Regulation (21 CFR part 820, incorporating ISO 13485:2016).

1. Define requirements (URS and FRS Part 1).
2. Assess risk with the FMEA (Part 2) and se the depth of testing for each requirement.
3. Approve this plan
4. Installation qualification (IQ): confirm the equipment is installed as specified.
5. Operational Qualification (OQ): Confirm every function works across its operating range, including failure and challenge tests (Part 4).
6. Performance Qualification (PQ): confirm consistent sterilization performance with real load.
7. Complete the traceability matrix (Part 5) and write the validation Summary Report.
8. Release for routine use, then maintain the validated state through change control and requalification.

Testing depth follows the FMEA: high risk gets scripted tests with challenge cases, medium risk gets scripted test, and Low risk get unscripted testing or review of supplier evidence.

Supplier factory and site acceptance test results may be used for low-risk items only after review and with written justification.

5. Roles and Responsibilities

Role	Responsibility
Validation Engineer	Writes and executes IQ, OQ, and PQ; records results; raises deviations
Validation Manager	Reviews and approves validation documents
QA	Approves the plan, protocols, deviations, and summary report
Operations	Runs Cycles under supervision; provides trained operators
IT/Administrator	Accounts, backup, restore, and network interface
Metrology	Calibration of instruments and test equipment
Supplier	Installation, factory and site acceptance test records, documentation

6. Deliverables

Document	Number	Purpose
URS and FRS	URS-FRS-1.1	What system must do and how (part 1)
FMEA and Risk assessment	RA- S101-001	Rates risk and set testing depth (part 2)
Validation Plan	VP-S101-001	This document (part 3)
Installation Qualification	IQ-S101-001	Installation checks (vendor provided manual guide)
Operational Qualifications	OQ-S101-001	Function and challenge test (Part 4)
Performance Qualifications	PA-S101-001	Loaded performance runs (not yet written)
Traceability Matrix	TM-S101-001	Link each URS to its FRS and tests (Part 5)
Validation summary Report	VSR-S101-001	Conclusion and release decision (Written after execution)

7. Test strategy and acceptance criteria

Phase	What is tested	Acceptance criteria
IQ	Documents, dimensions, weight, utilities, instrument calibration certificates, supplier test records	Every item matches the URS and supplier drawings

OQ	Security, audit trail, signatures, workflow controls, alarms, power failure, backup and restore, door lock, time synchronization, data recording, empty-chamber temperature mapping (3 consecutive cycles), air removal test, leak test	Every scripted steps passes; no open critical or major deviation
PQ	3 consecutive successful cycles of C-121 with load LC-01 and at least 12 temperature probes placed at the cold spots	Every probe stays between 121.0 C and 124.0 C during exposure; the lowest F0 is at least 15 minutes; no visible moisture; no alarms

A protocol step passes only when the actual result meets the expected result written before execution. A single failed step is a deviation.

8. Justification of key parameter values

Every example value in the URS and FRS needs a reason. Check each source before real use.

Parameter	Value
Sterilization setpoint	121.0C
Exposure time	20 minutes
Acceptance band	121.0 to 124.0C
Control band	121.0 to 123.0c
Minimum F0	15 minutes
Uniformity action limit	2.0C
Uniformity alert limit	1.5C
Leak rate	1.3 mbar per minute
Vacuum pulses	At least 3 to 70 mbar absolute or lower
Drying time	20 minutes of vacuum
Recording interval	Every 5 seconds
Alarm response	Within 2 seconds
Door lock pressure	0.05 bar
Password	8 or more characters with upper case, lower case, number, special character
Account lockout	5 failed logins
Inactivity logout	Dim at 3 minutes, log out at 5 minutes
Backup and restore	Every 24 hours, restore within 24 hours
UPS runtime	10 minutes
Availability	95% over 12 months
Requalification	At least annually and after significant change

9. Deviations

Any result that does not meet its expected result is recorded as deviation with a unique ID, its impact is assessed, and testing is repeated only after AQ approves the plan. Original results are never overwritten or removed. In this training project, deviations are logged in the DevTriage app.

10. Change control and requalification

Changes made before release, and any later change to hardware, software, recipes, or parameters, go through change control, which decides what to retest. S-101 is requalified at the interval above, and after any significant change (URS-2.1).

11. Documentation and data integrity

- Follow good documentation practice: blue or black in, date and initial every entry, correct mistakes with a single line and a reason, never overwrite.
- Record data so it is attributable, legible, contemporaneous, original, and accurate (ALCOA+).
- Using calibrated test equipment and record its ID and calibration due date in each protocol.
- Keep raw data (Printouts, probe files) with the executed protocol.

12. Traceability and reporting

The traceability matrix (Part 5) links every URS item to its FRS items and to the IQ, OQ, or PQ step that tests it. The validation summary report lists all deviations and their resolution, confirms that every acceptance criterion was met, and states whether S-101 is released for routine use.

13. Assumptions and limitations

- This is a training work sample with no physical sterilizer.
- The IQ, OQ, and PQ protocols are written but not executed. If any test is run on simulation or with synthetic data, the results must be clearly labeled SIMULATED.
- Never invent or fill in results that were not actually obtained. Fabricated results would be a serious data integrity violation.
- Example values in section 8 must be checked against current standards.

14. References

Reference	Used for

ISO 17665 (Check current edition)	Moist heat sterilization process and validation
EN 285 (Check current edition)	Large steam sterilizer performance values
21 CFR part 11	Electronic records and signatures
21 CFR part 820 (QMSR) with ISO 13485:2016	Quality System, calibration, process validation
ISPE GAMP 5 (Second Edition)	Risk-based computerized system validation
Parts 1 and 2 of these packages	Requirements and risk ratings

15. Approvals

Role	Name	Signature	Date
Prepared By (Validation Engineer)			
Reviewed By (Validation Manager)			
Approved By (QA)			